

Cross-Architecture Knowledge Distillation of Vision Transformers for Real-Time Multi-Label Chest X-ray Diagnosis on Embedded GPUs

Athanasios Georgousis, Eleni Politi, Theodoros Galetsas, Georgios Tsapikounis
School of Electrical and Computer Engineering
National Technical University of Athens (NTUA)
Athens, Greece

Abstract—Transferring representations between Vision Transformers (ViTs) and Convolutional Neural Networks (CNNs) remains an open challenge, driven by the mismatch between global self-attention and local convolutional receptive fields. The gap is acute in medical imaging, where CNNs are the only computationally viable option at the edge yet lack the global context that many diagnoses require. We present a cross-architecture knowledge distillation (KD) framework in which a ViT-B/16 teacher shapes a lightweight EfficientNet-B0 student ($21.3\times$ fewer parameters, $42.4\times$ fewer MACs), and we characterise the distilled student as a full deployment envelope rather than as a single accuracy number. On CheXpert, logit-level distillation reaches a macro AUROC of 0.8646 over the five competition pathologies, recovering 99.7% of the teacher’s 0.8675 ceiling and improving the undistilled baseline by $+0.069$ ($p < 10^{-4}$, paired bootstrap). A complete precision study (FP32/FP16/INT8) on an RTX 5070 Ti and a Jetson Orin Nano shows that FP16 is lossless, that INT8 tolerance is a property of soft-logit supervision *specifically* rather than of distillation in general, and that blur and quantization robustness are anti-correlated, mechanistically distinct axes. The Jetson attains ~ 70 img/s/W at INT8 and hosts the entire accuracy–energy Pareto frontier, with Logit-KD FP16 dominating at 22.9 mJ per image.

Index Terms—knowledge distillation, vision transformers, chest X-ray, CheXpert, edge AI, TensorRT, quantization, model compression

I. INTRODUCTION

Deep learning for medical imaging promises to reduce diagnostic fatigue and to surface micro-patterns imperceptible to the human eye. Computed tomography is a richer source of information, but it delivers far more radiation and is inaccessible to many smaller communities, so remote or underfunded regions stand to benefit most from cheap, high-accuracy evaluation of plain radiographs. Patient privacy adds a second constraint: localised hardware ensures data never leaves the hospital network. Edge AI satisfies both requirements, but introduces hard deployment constraints of its own.

A sharp architectural dichotomy governs the design space. Vision Transformers (ViTs) [1] dominate in accuracy by using self-attention [2] to capture global spatial relationships across an entire image, but their parameter count and memory footprint render them incompatible with low-power devices. CNNs, by contrast, are highly efficient and hardware-friendly, yet build context only hierarchically, through local convolutional receptive fields. The question this paper addresses is

therefore explicit: *can a compact CNN inherit the diagnostic behaviour of a ViT teacher and still run in real time, within a strict energy budget, on an embedded GPU?*

We answer it with a cross-architecture KD framework that distils a ~ 86 M-parameter ViT-B/16 into an EfficientNet-B0 student and carries the result end-to-end through ONNX/TensorRT compilation onto two hardware targets. Crucially, we treat the distilled model not as a point solution but as a *characterised operating envelope*. Our contributions are:

- A cross-architecture (ViT→CNN) distillation study over four students—no-KD baseline, logit-level, feature-level and feature+hard-label—in which logit-KD improves the baseline by $+0.069$ MACRO(5) AUROC on CheXpert and closes 99.7% of the teacher gap.
- A complete TensorRT precision study (FP32/FP16/INT8) on two devices, showing that FP16 is lossless and that INT8 tolerance is *specific to soft-logit supervision* rather than a generic property of distillation.
- The identification of a *robustness dissociation*: feature-level supervision buys blur robustness, logit-level supervision buys quantization robustness, and the two axes are anti-correlated.
- A full deployment envelope—accuracy–energy Pareto frontier, perf-per-watt, latency, thermal and cold-start behaviour—whose frontier is populated exclusively by Jetson configurations, together with a clinical-trustworthiness analysis (calibration, operating point) built on patient-level bootstrap confidence intervals.

II. RELATED WORK

Deep learning for chest X-rays. CNNs have historically been the backbone of radiograph analysis, and large-scale datasets such as CheXpert [3] and ChestX-ray14 [4] enabled radiologist-level results on individual findings [5]. However, CNNs excel at localised texture and lack the receptive field required for diffuse thoracic pathologies, which motivates transformer-based alternatives [6], [7].

Vision Transformers. ViTs process images as sequences of patches [1], [2], capturing long-range dependencies natively regardless of spatial distance; hierarchical [8] and dense-prediction [9] variants extend the paradigm further. Their computational cost, however, is prohibitive at the edge.

Knowledge distillation. Logit-level KD [10] trains a compact student to mimic the softened output distribution of a larger teacher, transferring implicit inter-class correlations; feature-level methods instead align intermediate representations [11], [12], and DeiT [13] shows distillation is effective inside the transformer family too (survey: [14]). The *cross-architecture* setting is substantially harder—patch tokens and convolutional feature maps are not directly comparable, so simple logit mimicking is often insufficient—and recent work introduces projection modules and attention bridges to span the structural gap [15], [16]. We adopt this line but, unlike prior work, evaluate it on the *deployed inference engines*, which is where the interaction between distillation and quantization becomes visible.

Efficient models and edge deployment. Compact backbones [17]–[20] and tiny-device pipelines [21] target constrained hardware, while post-training quantization [22] and graph-optimisation frameworks such as TensorRT [23] deliver the final latency and footprint reductions. Our work sits at the intersection of the two lines: we distil across architectures *and* push the result through the full quantization/compilation pipeline, then measure precisely what survives.

III. METHODOLOGY

A. Dataset and Preprocessing

CheXpert [3] provides 14 clinical observations with four label states (positive, negative, uncertain, missing). Missing entries are imputed as negative and all uncertain labels mapped to 0 (the U-Zeros policy), pushing the model towards definitive visual evidence and suppressing false positives at the cost of discarding diagnostic ambiguity. Because the dataset is severely skewed, imbalance is handled independently per class with $W_{pos} = N_{neg}/(N_{pos} + \epsilon)$, $\epsilon = 10^{-7}$, passed to BCEWithLogitsLoss for the teacher so that errors on rare pathologies are penalised more heavily. Radiographs are converted to RGB, resized to 224×224 , normalised, and augmented with random flips and slight rotations.

B. Teacher: ViT-B/16

The teacher is a ViT-B/16 initialised from a checkpoint already exposed to chest radiographs, with its classification head replaced by a linear layer mapping the 768-d representation to 14 outputs. It is fine-tuned for exactly 5 epochs with AdamW ($\text{lr} = 3 \times 10^{-5}$), cosine annealing and automatic mixed precision. The short, conservative schedule is deliberate: lacking the inductive biases of CNNs, ViTs overfit quickly on a single dataset, and starting from an intermediate checkpoint that has already learned foundational X-ray features makes a long schedule unnecessary. The final weights are frozen and serve as the teacher throughout (Appendix A).

C. Student and Distillation Variants

The student is EfficientNet-B0 [17]. Four variants are compared consistently throughout the paper.

(1) *Baseline.* Trained on hard labels only, with sigmoid focal loss [24] ($\gamma = 2.0$, $\alpha = 0.25$), which scales the loss

by prediction confidence and forces the optimiser onto hard, misclassified examples.

(2) *Logit-KD.* The student additionally matches the teacher’s 14 logits [10] under temperature scaling $T = 2.0$, with the two objectives balanced evenly ($\alpha_{KD} = 0.5$):

$$\mathcal{L}_{\text{total}} = (1 - \alpha_{KD}) \mathcal{L}_{\text{focal}} + \alpha_{KD} \mathcal{L}_{\text{soft}}. \quad (1)$$

(3) *Feature-KD.* A dual-projector bridge aligns the CNN features with the ViT token space [15]. Because the final EfficientNet block yields a 7×7 map—mismatched with the 196-token ViT-B/16 sequence—features are extracted from the second-to-last stage, which natively produces 14×14 maps and therefore flattens to exactly 196 elements, enabling one-to-one pixel-to-token alignment. A *partially cross-attention* projector lifts them into the 768-d space, with stochastic uniform masking ($p = 0.5$) element-wise substituting the student’s Query/Key/Value vectors with the teacher’s before a mixed attention map is computed—forcing the CNN to *structurally understand* the ViT’s attention distribution rather than copy it—while a *group-wise linear* projector aligns local receptive fields to the corresponding patches through grouped 1-D convolutions over 49 spatial neighbourhood groups. With $\mathcal{L}_{\text{Feat}} = \|\text{PCA}(F_S) - A_T\|_2^2 + \|\text{PCA}(V_S) - V_T\|_2^2 + \|\text{GL}(F_S) - F_T\|_2^2$, the objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{focal}} + \beta \mathcal{L}_{\text{Feat}}, \quad \beta = 0.01. \quad (2)$$

The small β is essential rather than cosmetic: at $\beta = 1$ the magnitude of the high-dimensional alignment loss overwhelms the gradients reaching the classification head (Appendix B).

(4) *Feature+Hard-KD.* Feature alignment is retained, but the teacher logits are removed entirely and the task loss reverts to hard labels. This variant isolates whether the benefits attributed to feature matching in fact require soft supervision—and, as Sec. V-E shows, calibration reveals that they do.

D. Edge Deployment Pipeline

Each student is exported to ONNX (opset 17) and compiled into device-specific TensorRT [23] engines with aggressive layer fusion and dynamic batch profiles ($\text{min}=1$, $\text{opt}=8$, $\text{max}=16$); compilation is architecture-specific, so engines are not transferable and are rebuilt on each target. Three engines per student are produced (FP32/FP16/INT8). INT8 needs special handling, since aggressive rounding collapses activation ranges: we use EntropyCalibrator2 with a *checkpoint-specific* calibration cache, built by passing a representative image subset through each network—reusing a cache across variants silently corrupts the mapped ranges. At runtime, CUDA memory is allocated for the 224×224 input and the 14-element output, and a host-side sigmoid yields the final multi-label probabilities.

IV. EXPERIMENTAL SETUP

A. Hardware and Software

Training and high-performance inference use an NVIDIA RTX 5070 Ti (Blackwell, sm_120, 12 GB) in a Lenovo

Legion workstation (Ubuntu 24.04, TensorRT 10.8). Edge deployment targets an NVIDIA Jetson Orin Nano Super (sm_87, JetPack 6.2, TensorRT 10.3). Inference is executed through a torch-free Polygraphy TrtRunner, which also verified the functional equivalence of feature maps across engines.

The two platforms expose power and memory through *entirely non-comparable* measurement domains: on the Legion, power is the GPU-package draw (`nvidia-smi`) and memory is on-GPU allocation; on the Jetson, power is the whole-module board rail `VDD_IN` (`tegrastats`) and memory is shared system RAM. All such figures are therefore interpreted *within* a device, to expose per-precision and per-variant trends, never as absolute cross-device comparisons—a caveat repeated in every affected caption.

B. Evaluation Data, Metrics and Statistical Protocol

All accuracy metrics are computed on the official CheXpert validation set of 234 studies, with patient identifiers retained for patient-level analysis. The primary metric is the macro-averaged AUROC over the five competition pathologies (Cardiomegaly, Edema, Consolidation, Atelectasis, Pleural Effusion), denoted MACRO(5). We deliberately do *not* headline MACRO(14): at $n=234$ several of the fourteen classes are severely under-represented—Fracture has no positive samples at all, yielding an undefined AUROC—which makes the 14-class mean unreliable.

Given the small validation set, significance is established with a *paired* bootstrap on the AUROC difference against the baseline (10,000 replicates [25]) rather than by comparing marginal intervals: resampling the same studies for both models cancels shared variance and tightens the interval on the difference, which is far more powerful at this sample size. We report image-level and patient-level results, the latter clustering by patient, the appropriate unit for medical data. A gain is significant when the 95% CI of Δ AUROC excludes zero.

C. Deployment and Robustness Protocols

Six protocols characterise deployment. (i) Perf-per-watt: throughput, board/GPU power, energy per image, and GPU-compute and end-to-end latency at batch 1. (ii) Thermal soak: a 600 s sustained-load run with per-second telemetry to detect throttling. (iii) Cold start and memory, measured under strict subprocess-per-engine isolation, so that each engine independently pays the one-time CUDA context-initialisation cost instead of amortising it across a shared process. (iv) Blur robustness: Gaussian blur at $\sigma \in \{0, 1, 2, 3, 4\}$ on the deployed engines. (v) Noise robustness at increasing standard deviations. (vi) Cross-dataset OOD generalisation on NIH ChestX-ray [4] over the classes shared with CheXpert. Full outputs and their interpretation are given in Appendices D–H.

V. RESULTS AND DISCUSSION

A. Distillation Efficacy

The ViT-B/16 teacher reaches a MACRO(5) ceiling of 0.8675, whereas the undistilled EfficientNet-B0 reaches 0.7954, trailing it by 0.072 (Table I). Distillation closes

TABLE I
MACRO(5) AUROC PER PRECISION AND DEVICE. THE TEACHER (ViT-B/16, FP32 PyTORCH) IS SHOWN AS THE ACCURACY CEILING.

Model	Legion			Jetson		
	FP32	FP16	INT8	FP32	FP16	INT8
ViT-B/16 (teacher)	0.8675	– (FP32 PyTorch ceiling)				
Baseline (no KD)	0.7954	0.7951	0.7275	0.7954	0.7953	0.7386
Logit-KD	0.8646	0.8646	0.8299	0.8644	0.8646	0.8327
Feature-KD	0.8577	0.8581	0.7859	0.8579	0.8580	0.7931
Feature+Hard-KD	0.8545	0.8543	0.7884	0.8542	0.8540	0.7956

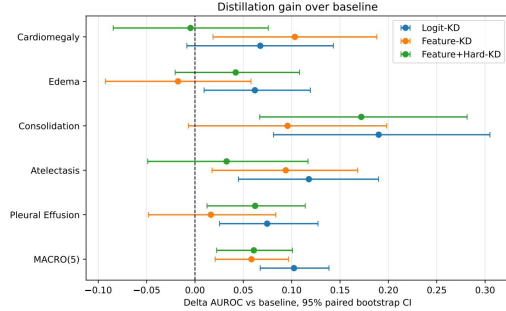


Fig. 1. Paired-bootstrap forest plot of Δ AUROC versus the baseline (Legion, INT8, image level, 10,000 replicates). Only Logit-KD retains a significant gain under quantization.

almost all of this gap: at FP16, Logit-KD attains 0.8646—within 0.0029 of the teacher, i.e. recovering $\sim 99.7\%$ of its score—a +0.0695 gain over the baseline. Feature-KD and Feature+Hard-KD also improve substantially (+0.0630 and +0.0592, reaching 0.8581 and 0.8543) but do not surpass Logit-KD. The two devices agree to four decimals at FP16, confirming that the result is a property of the distilled weights rather than of a deployment target. The gain is broad-based rather than driven by one easy class—Consolidation rises from 0.808 to 0.925, Pleural Effusion from 0.850 to 0.919, while Cardiomegaly improves least ($0.785 \rightarrow 0.810$)—as analysed in Appendix C.

1) *Statistical significance*: Because the validation set contains only 234 studies, every distillation gain is tested with the paired bootstrap of Sec. IV-B. On MACRO(5), all three distilled variants show a gain whose 95% CI excludes zero ($p < 10^{-4}$), at both image and patient level (Fig. 1). At the per-class level, the gain is significant for four of the five competition pathologies (Edema, Consolidation, Atelectasis, Pleural Effusion). We report openly that *Cardiomegaly is not significant* for any distilled variant—its interval includes zero at both levels—and treat this as an honest boundary of the method rather than omitting it: the smaller point gain is not statistically distinguishable from noise at this sample size. Patient-level intervals are wider, as expected, but preserve exactly the same pattern.

B. Precision and Quantization Robustness

FP16 is lossless. Across all variants and both devices, FP16 matches FP32 to within 0.0004 MACRO(5) while roughly

TABLE II

PERF-PER-WATT AT BATCH 1 (BASELINE AND LOGIT-KD; THE FEATURE-BASED VARIANTS ARE WITHIN NOISE, SEE APPENDIX D). **POWER DOMAINS ARE NOT DIRECTLY COMPARABLE: LEGION = GPU-PACKAGE (NVIDIA-SMI); JETSON = WHOLE-MODULE BOARD (TEGRASTATS).**

Variant	Prec.	Legion			Jetson		
		img/s	mJ/img	img/s/W	img/s	mJ/img	img/s/W
Baseline	FP32	1049	66.70	15.0	338	43.00	23.3
	FP16	1858	37.43	26.7	471	23.36	42.8
	INT8	1504	46.53	21.5	621	14.36	69.6
Logit-KD	FP32	1069	65.41	15.3	338	42.03	23.8
	FP16	1821	38.13	26.2	467	22.87	43.7
	INT8	1521	45.98	21.7	617	14.28	70.0

halving the engine footprint (the Logit-KD engine drops from 15.6MB to 8.8MB); it is therefore the default deployment precision, at no accuracy cost.

INT8 exposes a variant-specific gap. Aggressive 8-bit quantization degrades accuracy, but not uniformly. The drop is far smaller for Logit-KD ($0.8646 \rightarrow 0.8299$ on Legion and $0.8646 \rightarrow 0.8327$ on Jetson, i.e. ~ 0.032 – 0.035) than for the feature-based variants (~ 0.07). We attribute this to the smoother output distributions produced by soft-target supervision, which present a narrower, more quantization-friendly activation range to the INT8 calibrator. Quantization robustness is thus a specific property of *logit-level* distillation, not a generic consequence of distillation—a distinction that, to our knowledge, has not been made explicit in prior work.

The Logit-KD advantage widens under quantization. Its margin over the baseline grows from $+0.0693$ at FP16 to $+0.0941$ at INT8 on the Jetson, because the baseline degrades faster than the soft-target student. The INT8 ranking is identical on two independently calibrated hardware generations (TensorRT 10.8 vs. 10.3), indicating a real property of the trained distributions rather than a calibration artefact; the slight cross-device difference ($\Delta \approx 0.011$) reflects independent per-device calibration, not hardware superiority.

C. Efficiency and the Accuracy–Energy Frontier

Accuracy alone does not establish deployability. Table II exposes a clean device-specific split at batch 1. On the Legion, package power is flat across precisions (69.4–70.0 W), so efficiency is governed entirely by throughput and FP16 is the best operating point (~ 26 img/s/W). On the Jetson, board power scales down with precision (14.2 W at FP32 to 8.8 W at INT8) and INT8 dominates at ~ 70 img/s/W—roughly $3\times$ the FP32 efficiency—driven by the Orin INT8 tensor cores. The ordering is invariant to the distillation variant: per-per-watt is a property of the architecture–precision–hardware combination, not of the training objective.

The Pareto frontier lies entirely on the edge device. Fig. 2 plots MACRO(5) against energy per image for all (device, variant, precision) configurations. The globally non-dominated frontier consists *exclusively of Jetson points*: at comparable accuracy the Legion spends far more energy per inference (~ 38 mJ versus 22.9 mJ for the same 0.8646

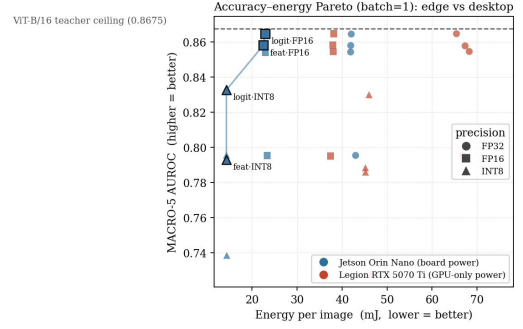


Fig. 2. Accuracy–energy frontier over all (device, variant, precision) configurations; the non-dominated frontier is exclusively Jetson. Energy domains are per-device and not cross-comparable.

AUROC), its fixed ~ 70 W package power dominating the per-image cost despite $\sim 4\times$ higher raw throughput—and the comparison is deliberately stacked *against* the Jetson, whose figure is whole-board energy while the Legion’s is GPU-only. The cause is a scale mismatch, not a defect of the method: a 0.40 GMAC student severely underutilises a desktop-class GPU that still draws its large static power. Along the Jetson frontier, **Logit-KD FP16 is the recommended operating point** (0.8646 at 22.9 mJ per image), with its INT8 counterpart (~ 14.3 mJ at 0.833) valid whenever a ~ 0.03 AUROC cost buys a further $\sim 38\%$ energy reduction. Efficient edge inference thus comes from matching a compact, distilled model to a right-sized accelerator, not from maximising raw compute; Appendix E develops the frontier and the elastic-deployment argument that follows from it.

Latency and the precision inversion. Both devices sustain real-time single-image latency, but precision affects them oppositely. On the Jetson the expected ordering holds and INT8 is fastest (617 versus 467 img/s for Logit-KD), its dedicated INT8 tensor cores repaying the quantize/dequantize overhead. On the Legion the ordering *inverts*—INT8 is *slower* than FP16 (1521 versus 1821 img/s)—because for a model this compact that fixed overhead is never amortised at batch 1 on a much larger GPU. Precision must therefore be selected per device, never globally (Appendix D).

D. Deployment Reliability and the Robustness Dissociation

Evaluating the deployed engines under Gaussian blur reveals a *dissociation between clean accuracy and blur robustness*. At FP16, Feature-KD is by far the most blur-robust variant (area under the degradation curve, AUD ≈ 0.256 ; retention $\approx 79\%$) despite not having the highest clean score, whereas Logit-KD—the strongest clean and most quantization-robust variant—degrades the fastest (AUD ≈ 0.694 ; retention $\approx 54\%$), below even the baseline ($\approx 66\%$). Fig. 3 makes the structure explicit: plotting quantization robustness ($\text{AUROC}_{\text{INT8}}/\text{AUROC}_{\text{FP16}}$) against blur robustness ($\text{AUROC}_{\sigma=4}/\text{AUROC}_{\sigma=0}$), Logit-KD occupies the quant-robust/blur-fragile quadrant ($\sim 96\%$ versus $\sim 54\%$)

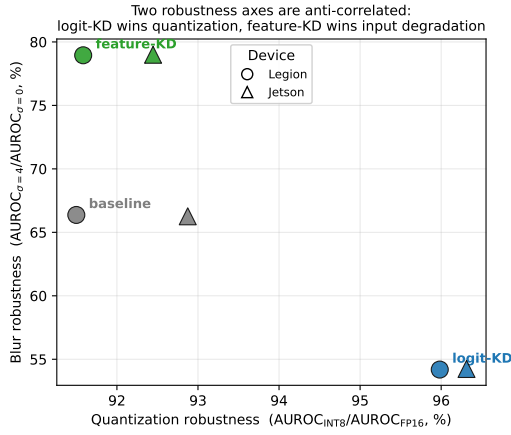


Fig. 3. Quantization robustness versus blur robustness. The two axes are anti-correlated: Logit-KD is quant-robust/blur-fragile, Feature-KD is blur-robust/quant-fragile. The “robust to both” quadrant is empty.

and Feature-KD the blur-robust/quant-fragile quadrant ($\sim 92\%$ versus $\sim 79\%$), on both devices.

The two axes are therefore *anti-correlated and mechanistically distinct*: blur robustness reflects the preservation of spatial representations—exactly what feature-level alignment to the ViT’s 14×14 token grid reinforces—whereas quantization robustness reflects the smoothness of the output distribution, which soft-logit supervision produces. The “robust to both” corner of Fig. 3 is empty, and reaching it is an open methodological problem rather than a matter of naive combination: Feature-KD already couples feature alignment with soft logits, yet still lands in one quadrant only (Appendix G).

Thermal and cold-start behaviour. A 600 s sustained-load soak throttles neither device: throughput retention is $\geq 99.4\%$ in all conditions, with the Legion stabilising at $\sim 60^\circ\text{C}$ and the Jetson markedly cooler at $\sim 45^\circ\text{C}$. Cold activation is uniform across engines ($\sim 0.18\text{--}0.22$ s, invariant to the distillation variant) and Jetson peak memory is modest and precision-ordered (189–279 MB of system RAM, under 4% of the 8 GB module). Sustained real-time operation is therefore thermally safe without active intervention, and models can be instantiated on demand rather than pinned resident—an elastic-deployment property analysed in Appendices E and F.

E. Clinical Trustworthiness

AUROC quantifies ranking alone. For a compressed model intended for edge screening, two further properties determine whether its outputs are clinically actionable: whether the emitted probabilities are trustworthy (*calibration* [26]), and how the model behaves at a concrete decision threshold, where the cost of a missed finding is paramount.

Calibration. At FP16, soft-target distillation roughly halves the baseline Expected Calibration Error (Table III: ECE $0.123 \rightarrow 0.084$ for Logit-KD, 0.081 for Feature-KD). The benefit is soft-target-specific: Feature+Hard-KD, which discards the teacher logits while retaining feature matching, does *not* improve calibration (ECE 0.133), locating the gain in soft

TABLE III
CLINICAL TRUSTWORTHINESS ON THE JETSON. LEFT: CALIBRATION OVER THE FIVE COMPETITION CLASSES (MACRO-5, LOWER IS BETTER). RIGHT: SENSITIVITY AND SPECIFICITY AT YODEN’S J , FP16. LEGION FP16 VALUES ARE WITHIN NUMERICAL TOLERANCE (FP16 LOSSLESSNESS), WHICH IS AN ACCURACY-DOMAIN EQUIVALENCE AND *not* A CROSS-DEVICE POWER OR MEMORY COMPARISON.

Variant	FP16		INT8		FP16 @ Youden	
	ECE	Brier	ECE	Brier	Sens.	Spec.
Baseline	0.123	0.163	0.107	0.172	0.769	0.729
Logit-KD	0.084	0.129	0.119	0.151	0.867	0.754
Feature-KD	0.081	0.132	0.186	0.182	–	–
Feature+Hard-KD	0.133	0.153	0.104	0.164	–	–

supervision rather than feature alignment per se. Under INT8 the degradation is selective—Feature-KD collapses ($0.081 \rightarrow 0.186$) whereas Logit-KD degrades only mildly ($0.084 \rightarrow 0.119$)—mirroring the AUROC behaviour of Sec. V-B along the calibration axis.

Operating point. Selecting, per pathology, the threshold that maximises Youden’s J [27], Logit-KD raises sensitivity on *all five* findings, lifting mean sensitivity from 0.769 to 0.867 —an ≈ 0.10 absolute reduction in missed findings—most markedly for Consolidation ($0.818 \rightarrow 0.939$). Critically this is not a mere threshold shift towards recall: mean specificity also rises ($0.729 \rightarrow 0.754$), so the reduction in false negatives does not, on average, cost additional false positives. The sole exception is Cardiomegaly, whose sensitivity gain ($0.750 \rightarrow 0.882$) is paid for with specificity ($0.747 \rightarrow 0.627$), consistent with its non-significant AUROC delta. Three grounded factors explain this: it is adjudicated from the cardiothoracic ratio, a global geometric relationship that a globally pooled CNN already captures, leaving little exploitable “dark knowledge” in the soft targets; its labels are NLP-derived and projection-dependent, injecting noise that caps the achievable AUROC; and with only 68 positives the bootstrap is underpowered (Appendix H). For sensitivity-prioritising screening the trade is defensible, but we report it explicitly rather than claiming a uniform gain.

F. Robustness Inheritance and Qualitative Interpretability

Conventionally trained CNNs decide largely on texture and degrade sharply once it is removed [28], whereas ViTs are comparatively robust. Across progressively corrupted inputs, the more ViT supervision a student receives the more of this robustness it inherits, and the ordering reappears under domain shift: on NIH ChestX-ray [4], acquired with different equipment, the feature+logit student retains 93.7% of its in-distribution AUROC over the eight shared classes versus 89.8% for the baseline (Appendix G).

Grad-CAM [29] asks *where* each model looks. We focus on Pleural Effusion, whose signature is anatomically well defined: gravity-dependent fluid blunts the costophrenic angle and produces a basal, meniscus-shaped opacity. Saliency is extracted at each model’s native representation—the teacher’s final pre-attention LayerNorm with the 14×14 token grid reshaped to image space, the students’ final 7×7 convolutional map—so Fig. 4 contrasts different mechanisms at different resolutions.

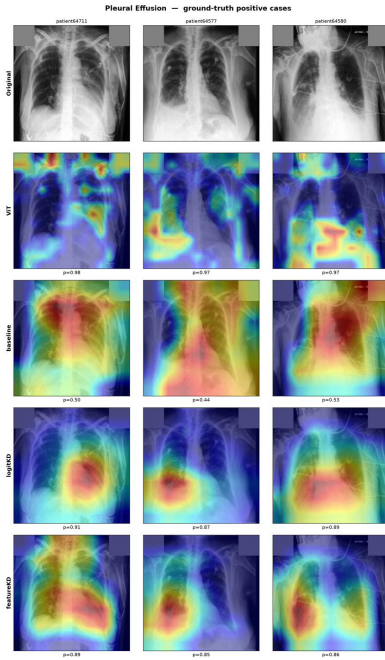


Fig. 4. Grad-CAM on ground-truth-positive Pleural Effusion cases: ViT-B/16 teacher, Baseline, Logit-KD, Feature-KD. Qualitative and illustrative only.

We therefore neither expect nor claim that distillation transfers the teacher’s *spatial* attention: the teacher panel exposes the mechanism gap, not attention alignment. What transfers is *predictive* behaviour at the logit level. The fair comparison is student-versus-student, where identical architecture and target layer make any difference attributable to the training signal alone: read anatomically, the distilled students concentrate saliency on the lower lung zones and costophrenic angles more consistently than the baseline, whose maps are diffuse. This is hypothesis-consistent with the sensitivity gains of Sec. V-E, not a measurement: lacking region-level annotations we report no localisation score (Appendix H).

VI. LIMITATIONS

The official CheXpert validation set contains only 234 studies, so per-class positive counts are scarce, AUROC estimates carry substantial variance, and Fracture has no positives at all. This destabilises the 14-class macro average—hence MACRO(5) as the primary metric—and limits statistical power, which is precisely why the Cardiomegaly gain does not reach significance. External evaluation was limited to a single out-of-distribution cohort (NIH ChestX-ray) of retrospectively assessed frontal radiographs: it probes generalisation but does not establish clinical validity across institutions, protocols, demographics or prevalence, and no multi-centre or prospective evaluation was performed.

The framework is inherently bounded by its teacher—any bias, calibration error or diagnostic weakness of the ViT propagates to the student, whose performance is capped by a teacher that itself plateaued within a short schedule—and by the label policy, since mapping missing and uncertain labels to

the negative class suppressed false positives but discarded diagnostic uncertainty that an uncertainty-aware objective might preserve.

Deployment characterisation was restricted to NVIDIA hardware and the TensorRT runtime, so latency, throughput and quantization behaviour may not transfer to other accelerators or frameworks, and because the two platforms expose power and memory through different domains, cross-device analysis is confined to throughput ratios. Reduced precision is likewise not uniformly beneficial: INT8 improves throughput and energy but degrades accuracy and calibration in a variant-dependent way—only the logit-distilled student remains robust—so it is advisable only for schemes whose output distributions remain quantization-tolerant. Finally, the interpretability analysis is qualitative, the feature-level variants add projection and attention-masking modules that raise training cost, and a systematic ablation of the distillation hyperparameters was left outside the present scope.

VII. CONCLUSION AND FUTURE WORK

We presented a cross-architecture knowledge distillation framework that transfers the diagnostic behaviour of a ViT-B/16 teacher to a lightweight EfficientNet-B0 student for multi-label chest X-ray classification, carried end-to-end through TensorRT compilation onto an embedded GPU. The distilled students significantly outperform the undistilled CNN—Logit-KD recovers 99.7% of the teacher’s MACRO(5) at $21.3\times$ fewer parameters—while inheriting properties absent from the baseline: better calibration, higher sensitivity across all five competition findings, and greater robustness to corruption and domain shift. Beyond accuracy, the work delivers a characterised deployment envelope: FP16 is lossless; quantization tolerance is a specific consequence of soft-logit supervision; blur and quantization robustness are anti-correlated, mechanistically distinct axes; and the entire accuracy–energy Pareto frontier lies on the edge device, with Logit-KD FP16 dominating at 22.9 mJ per image.

This characterisation is *task-agnostic and extensible*: the pipeline delivers a reproducible route rather than a classifier, with two orthogonal degrees of freedom—accuracy–energy along the Pareto frontier, latency–throughput along the batch sweep. Because the weights are fixed and only the operating point moves, the same artefact scales from latency-bound single-stream to throughput-bound batched operation without retraining; combined with sub-second cold start and a small memory footprint, this supports an elastic “one edge node, many diagnostic models” deployment (Appendix E). Future work targets the empty “robust to both” quadrant of Fig. 3—via quantization-aware training, annealed loss weighting, or staged distillation—external multi-institutional validation with a second student architecture, and uncertainty-aware objectives. Prospective clinical evaluation remains necessary before safety-critical deployment.

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APPENDIX A TEACHER FINE-TUNING TRAJECTORY

The teacher determines the ceiling of the entire study, so its training was kept deliberately short and was monitored per epoch. Over five epochs the training loss fell steadily from 0.9415 to 0.7934, while the validation loss settled at 1.4658—the point at which the model’s usable capacity for this dataset had been reached, and beyond which a transformer without convolutional inductive biases would begin to overfit the $\sim 224\text{k}$ -image training split rather than generalise from it. Because the network was initialised from an intermediate checkpoint that had already learned foundational X-ray features, this short schedule was sufficient to adapt the $\sim 86\text{M}$ -parameter backbone to the CheXpert task without degrading its generalisation.

Table IV reports the per-epoch, per-class validation AUROC. Two patterns matter for the rest of the paper. First, the majority classes are learned reliably and early—Pleural Effusion stabilises around 0.92 and Consolidation around 0.91 by epoch 2—which is why these classes are also where the student inherits the largest gains. Second, the rare classes are erratic and, in several cases, actively misleading: Lung Lesion oscillates between 0.155 and 0.348, Enlarged Cardiomeastinum drops to 0.468 by epoch 5, and Fracture is undefined throughout because the validation set contains no positive samples for it. Pleural Other reaching a nominal 1.000 at epoch 5 is not a genuine perfect classifier but an artefact of a handful of positives. These observations are the empirical justification for adopting MACRO(5) as the primary metric: the 14-class mean is dominated by classes whose AUROC estimates are effectively noise at $n=234$.

TABLE IV
ViT-B/16 TEACHER VALIDATION AUROC ACROSS ALL FIVE FINE-TUNING EPOCHS. FRACTURE IS UNDEFINED (NO POSITIVE SAMPLES IN THE VALIDATION SET).

Pathology	Ep. 1	Ep. 2	Ep. 3	Ep. 4	Ep. 5
No Finding	0.880	0.890	0.875	0.865	0.890
Enlarged Cardiomeastinum	0.576	0.577	0.545	0.595	0.468
Cardiomegaly	0.832	0.766	0.772	0.794	0.769
Lung Opacity	0.904	0.898	0.902	0.893	0.884
Lung Lesion	0.180	0.155	0.292	0.348	0.228
Edema	0.906	0.909	0.912	0.923	0.904
Consolidation	0.915	0.919	0.904	0.898	0.914
Pneumonia	0.589	0.677	0.622	0.667	0.679
Atelectasis	0.779	0.786	0.779	0.769	0.792
Pneumothorax	0.795	0.808	0.838	0.876	0.851
Pleural Effusion	0.919	0.927	0.920	0.926	0.927
Pleural Other	0.961	0.991	0.940	0.996	1.000
Fracture	N/A	N/A	N/A	N/A	N/A
Support Devices	0.813	0.819	0.836	0.824	0.848

APPENDIX B

FEATURE-DISTILLATION DESIGN AND THE β CONSTRAINT

The feature-level pipeline required one non-obvious design decision, and it is worth stating explicitly because it is the difference between the method working and failing outright.

In the initial configuration the feature-distillation loss was weighted equally with the task classification loss ($\beta = 1.0$). This proved strongly suboptimal. $\mathcal{L}_{\text{Feat}}$ is an ℓ_2 loss computed over a 196×768 representation, so its magnitude is orders of magnitude larger than that of the focal loss computed over 14 logits. During backpropagation the alignment gradients therefore dominated the shared backbone and effectively starved the classification head, and task-specific accuracy degraded even as the student’s features became more ViT-like. The pathology is instructive: the student was successfully learning to *imitate* the teacher’s representation while simultaneously losing the ability to *use* it.

Constraining the coefficient to $\beta = 0.01$ resolves the bottleneck. At this scale the feature term acts as an *architectural regulariser*—it biases the CNN’s intermediate representation towards the ViT’s spatial priors—without distorting the decision boundary that the focal loss is shaping. In the Feature+Hard-KD variant we additionally isolate the task loss entirely by removing the teacher logits and relying strictly on ground-truth hard labels; this is the ablation that later reveals, through calibration (Sec. V-E), that the calibration benefit of distillation comes from the soft targets and not from feature alignment.

For completeness, the combined feature-plus-logit objective evaluated in the paper is

$$\mathcal{L}_{\text{total}} = \underbrace{(1 - \alpha)\mathcal{L}_{\text{focal}} + \alpha\mathcal{L}_{\text{soft}}}_{\mathcal{L}_{\text{task}}} + \beta\mathcal{L}_{\text{Feat}}, \quad (3)$$

with $\alpha = 0.5$, $T = 2.0$ and $\beta = 0.01$ throughout.

APPENDIX C

PER-CLASS ACCURACY, ABLATIONS AND MODEL COMPLEXITY

A. What each distillation signal actually recovers

Table V gives the checkpoint-level per-class validation AUROC of all four students and is the evidence behind the claim, in Sec. V-A, that the distillation gain is broad-based rather than concentrated in one easy class.

The baseline’s failure modes are architectural, not incidental. Its worst results—Pneumothorax at 0.672 and Consolidation at 0.808—are precisely the pathologies whose radiographic evidence is diffuse or requires comparing distant regions of the lung field, which a purely local convolutional receptive field cannot integrate. Logit-KD lifts exactly these: Consolidation rises to 0.925 and Pleural Effusion to 0.919. This is consistent with the mechanism we claim—mimicking the teacher’s continuous output distribution transfers implicit geometric priors about patch proximity and global context that hard clinical annotations obscure.

The feature-based variants recover something different. Pure Feature-KD reaches a 14-class macro of 0.775, *above* Logit-KD’s 0.768, despite a lower MACRO(5), because its advantage is concentrated in the sparse classes: Pleural Other surges to 0.936 (baseline 0.412). Forcing regional convolutional receptive fields to structurally match transformer hidden dimensions appears to reinforce contextual definitions where the label signal alone is too weak to do so. The clearest case is Lung Lesion, where the baseline collapses entirely to a catastrophic 0.047—worse than random, i.e. an inverted ranking—and the combined feature+logit scheme lifts it to 0.298. This remains poor in absolute terms, but the direction of the effect is unambiguous: multi-group structural regularisation prevents the gradients from flattening on highly isolated, rare abnormalities.

We stress that these rare-class figures are reported for mechanistic insight only and are *not* defensible headline numbers. With a handful of positives each, their AUROC estimates are unstable, which is exactly why MACRO(5) is the primary metric of the paper.

TABLE V
PATHOLOGY-WISE EPOCH-5 VALIDATION AUROC ACROSS DISTILLATION STRATEGIES (PYTORCH CHECKPOINTS, FP32).

Pathology	Baseline	Logit-KD	Feature+Logit	Feature-only
No Finding	0.841	0.879	0.872	0.880
Enlarged Cardiomediastinum	0.564	0.618	0.622	0.576
Cardiomegaly*	0.785	0.810	0.812	0.785
Lung Opacity	0.842	0.893	0.901	0.887
Lung Lesion	0.047	0.107	0.107	0.298
Edema*	0.847	0.899	0.894	0.907
Consolidation*	0.808	0.925	0.892	0.910
Pneumonia	0.483	0.640	0.639	0.730
Atelectasis*	0.688	0.769	0.787	0.764
Pneumothorax	0.672	0.770	0.743	0.708
Pleural Effusion*	0.850	0.919	0.906	0.906
Pleural Other	0.412	0.901	0.936	0.854
Fracture	N/A	N/A	N/A	N/A
Support Devices	0.780	0.849	0.842	0.871
Macro AUROC (14)	0.663	0.768	0.766	0.775
Macro AUROC (5-comp*)	0.795	0.864	0.858	0.854

*Core 5-competition CheXpert subset.

B. Compression accounting

The ViT-B/16 teacher has 85.81 M parameters and 16.87 GMACs; the EfficientNet-B0 student has 4.03 M parameters and 0.40 GMACs—a $21.3\times$ reduction in parameters and $42.4\times$ in multiply-accumulate operations.¹ On disk this compression compounds with reduced-precision serialisation: the teacher checkpoint occupies 327.4 MB, while the distilled student shrinks from 15.6 MB (FP32 checkpoint) to 8.8 MB (FP16 engine) and 7.0 MB (INT8 engine)—roughly $47\times$ smaller than the teacher.

Table VI confirms an important negative result: engine size is *invariant* to the distillation variant, varying only with precision and build target (± 0.13 MB across all four students at a given precision). Distillation changes the weights, not the graph, so none of the accuracy or robustness differences reported in this paper can be attributed to a difference in

¹The 4.03 M figure reflects the 14-output classification head used here; the commonly cited 5.3 M corresponds to the ImageNet-1000 configuration of EfficientNet-B0.

deployed model size. The Jetson’s smaller INT8 engines (~ 5.8 MB versus ~ 7.0 MB) reflect a different TensorRT version and kernel-selection policy, not a different model.

TABLE VI
ON-DISK TENSORRT ENGINE FOOTPRINT (MB), BATCH 1. SIZE IS VARIANT-INVARIANT; IT DEPENDS ONLY ON PRECISION AND BUILD TARGET.

Variant	Legion			Jetson		
	FP32	FP16	INT8	FP32	FP16	INT8
Baseline	16.96	8.86	7.04	16.89	9.31	5.77
Logit-KD	16.98	8.84	7.04	16.67	9.30	5.82
Feature-KD	17.00	8.89	7.02	16.69	9.30	5.79
Feature+Hard-KD	17.09	8.95	7.05	16.91	9.31	5.82

APPENDIX D FULL EFFICIENCY, LATENCY AND THROUGHPUT CHARACTERISATION

A. Perf-per-watt across all variants

Table VII extends the two-variant summary of the main body to all four students, and Fig. 5 plots the same data. The spread across variants at a fixed (device, precision) cell is at most $\sim 3\%$ in throughput and ~ 1 img/s/W in efficiency—within run-to-run noise. This is the expected result and it is worth stating as a positive finding rather than a null one: because all four students share the EfficientNet-B0 graph, the distillation objective changes *what the weights encode* but not *what the hardware executes*. Every efficiency conclusion in this paper is therefore a statement about the architecture–precision–hardware combination, and the choice of distillation signal can be made purely on accuracy, robustness and calibration grounds, at zero deployment cost.

The device-specific split is the substantive finding. Legion package power is pinned at ~ 70 W regardless of precision, so its efficiency curve simply follows throughput and peaks at FP16. Jetson board power falls monotonically with precision ($14.2 \rightarrow 10.7 \rightarrow 8.8$ W), so INT8 wins twice—it is both faster and cheaper—and reaches ~ 70 img/s/W, roughly triple its own FP32 figure. **The two power columns are not cross-comparable:** Legion reports GPU-package power via `nvidia-smi`, Jetson reports whole-module board power (VDD_IN) via `tegrastats`.

TABLE VII
PERF-PER-WATT AT BATCH 1, ALL FOUR VARIANTS. POWER DOMAINS ARE PER-DEVICE AND *not* CROSS-COMPARABLE.

Variant	Prec.	Legion				Jetson			
		img/s	W	mJ	img/s/W	img/s	W	mJ	img/s/W
Baseline	FP32	1049	70.0	66.70	15.0	338	14.5	43.00	23.3
	FP16	1858	69.6	37.43	26.7	471	11.0	23.36	42.8
	INT8	1504	70.0	46.53	21.5	621	8.9	14.36	69.6
Logit-KD	FP32	1069	69.9	65.41	15.3	338	14.2	42.03	23.8
	FP16	1821	69.4	38.13	26.2	467	10.7	22.87	43.7
	INT8	1521	70.0	45.98	21.7	617	8.8	14.28	70.0
Feature-KD	FP32	1039	70.0	67.34	14.9	338	14.2	41.92	23.9
	FP16	1830	69.5	37.96	26.3	475	10.7	22.47	44.5
	INT8	1548	70.0	45.20	22.1	616	8.8	14.28	70.0
Feat+Hard	FP32	1025	69.9	68.26	14.7	339	14.2	41.88	23.9
	FP16	1827	69.5	38.05	26.3	470	10.8	22.95	43.6
	INT8	1549	70.0	45.18	22.1	617	8.8	14.29	70.0

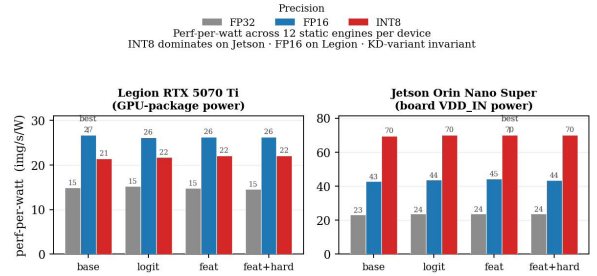


Fig. 5. Perf-per-watt by device, variant and precision. FP16 is optimal on the Legion, INT8 on the Jetson; the ordering is invariant to the distillation variant. Power domains are per-device and not cross-comparable.

B. Latency and the precision inversion

Table VIII reports GPU-compute time, end-to-end latency and engine-load time at batch 1. Both platforms sustain real-time single-image inference—sub-millisecond GPU time on the Legion, 1.6–3.0 ms on the Jetson—but the effect of precision inverts between them.

On the Jetson, INT8 is the fastest engine (~ 1.62 ms versus ~ 2.13 ms at FP16), because the Orin’s dedicated INT8 tensor cores more than repay the quantize/dequantize overhead. On the Legion, INT8 is *slower* than FP16 (~ 0.66 ms versus ~ 0.55 ms). The explanation is a workload–hardware mismatch: for a 0.40 GMAC model at batch 1, the quantize/dequantize cost is a fixed, non-amortisable overhead, while the theoretical INT8 compute advantage requires enough arithmetic work to be realised. A larger GPU simply finishes the FP16 arithmetic before the INT8 overhead can pay for itself. This is the latency-domain counterpart of the perf-per-watt split, and it has a direct practical consequence: precision should be selected per device, never globally.

The one anomalous entry deserves an explicit note. The 2.658 s engine-load time in the Baseline/FP16 Legion cell is not a per-engine cost but a one-time GPU driver and CUDA-context initialisation paid by whichever engine happens to run first in a fresh session. All other Legion loads cluster tightly at ~ 0.18 s. We exclude it from per-engine reporting and, in

the cold-start protocol (Appendix E), eliminate it structurally by isolating every engine in its own subprocess.

TABLE VIII
DEPLOYMENT LATENCY AT BATCH 1: GPU-COMPUTE TIME (HARNES-DEPENDENT), END-TO-END LATENCY, AND ENGINE LOAD TIME. THE 2.658 s ENTRY IS A ONE-TIME GPU DRIVER INITIALISATION, NOT A PER-ENGINE COST.

Variant	Prec.	Legion			Jetson		
		GPU (ms)	e2e (ms)	load (s)	GPU (ms)	e2e (ms)	load (s)
Baseline	FP32	0.966	0.972	0.183	2.958	3.021	0.303
	FP16	0.545	0.551	2.658	2.120	2.180	0.050
	INT8	0.670	0.676	0.183	1.605	1.655	0.137
Logit-KD	FP32	0.936	0.942	0.179	2.953	3.017	0.292
	FP16	0.552	0.558	0.177	2.136	2.196	0.220
	INT8	0.663	0.668	0.179	1.616	1.669	0.129
Feature-KD	FP32	0.965	0.972	0.180	2.951	3.016	0.296
	FP16	0.548	0.553	0.181	2.103	2.165	0.210
	INT8	0.652	0.658	0.181	1.620	1.670	0.130
Feat+Hard	FP32	0.978	0.985	0.183	2.950	3.013	0.301
	FP16	0.556	0.561	0.181	2.123	2.183	0.207
	INT8	0.687	0.728	0.178	1.617	1.670	0.130

C. Throughput scaling with batch size

Fig. 6 sweeps the Jetson batch size from 1 to 16 using the dynamic engine. Throughput rises with batch size while per-batch latency grows sub-linearly, so applications that can tolerate batching realise substantially higher sustained throughput than the batch-1 figures of Table VII suggest. This exposes a second degree of freedom that is *orthogonal* to the accuracy–energy trade-off: latency versus throughput. The same compiled engine can therefore serve a strictly latency-bound single-stream clinical workstation and a throughput-bound batch-screening queue, with no retraining and no re-export—only a change of operating point. This is the concrete mechanism behind the scalability argument of Appendix E.

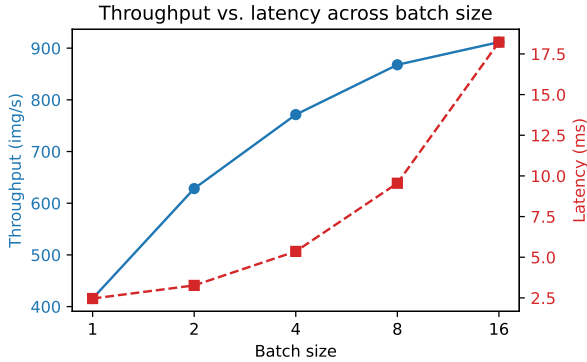


Fig. 6. Jetson batch-size sweep (1–16) with the dynamic engine. Throughput rises while per-batch latency grows sub-linearly.

APPENDIX E

THE ACCURACY–ENERGY FRONTIER, COLD START, AND THE SCALABILITY OF THE EDGE NODE

This appendix develops the argument that the contribution of the pipeline is a *characterised envelope* rather than a point solution, by combining three results: where the Pareto frontier

lies, how cheaply a model can be started and held in memory, and what those two facts jointly permit.

A. The accuracy–energy Pareto frontier

Fig. 2 plots MACRO(5) AUROC against energy per image for all (device, variant, precision) configurations, with the teacher ceiling as a reference line. The globally non-dominated frontier—the set of configurations that cannot be improved in accuracy without spending more energy, or in energy without losing accuracy—consists *exclusively of Jetson points*. Not a single Legion configuration is Pareto-optimal.

This is not a statement that the Legion is slow. In raw throughput it processes roughly $4\times$ more images per second than the Jetson at FP16 (~ 1821 versus ~ 467 img/s for Logit-KD). But on every deployment-relevant efficiency metric the Jetson wins decisively: its best perf-per-watt (~ 70 img/s/W) is nearly triple the Legion’s best (~ 26 img/s/W), and its energy per image is lower even under the most conservative possible comparison. That last point deserves emphasis, because the measurement domains are stacked *against* the Jetson: its 22.9 mJ figure is whole-module board energy, including CPU, memory and every peripheral, whereas the Legion’s 38.1 mJ is GPU-package energy only. The true system-level gap is therefore larger than the numbers suggest.

The cause is a scale mismatch between workload and hardware, not a defect of the method. A 4.03 M-parameter, 0.40 GMAC student severely underutilises a desktop-class GPU: most streaming multiprocessors sit idle while the device still draws its large static package power, so high throughput is bought at a fixed power floor that the tiny workload cannot amortise. The Jetson, engineered for exactly this power envelope, keeps the accelerator well-matched to the workload and pays no comparable static tax. The frontier therefore reframes the result as evidence *for* the central thesis of the paper: efficient edge inference comes from matching a compact, distilled model to a right-sized accelerator, not from maximising raw compute.

Along the Jetson frontier the choice is principled rather than ad hoc. Logit-KD FP16 is the recommended sweet spot: 0.8646 MACRO(5)—within 0.003 of the teacher ceiling—at only 22.9 mJ per image. Its INT8 counterpart sits lower and to the left (~ 14.3 mJ at 0.833 MACRO(5)), a legitimate frontier point whenever the ~ 0.03 AUROC cost is an acceptable price for a further $\sim 38\%$ energy reduction. An energy budget thus maps directly onto a configuration.

B. Cold-start latency and memory footprint

Cold-start latency and memory footprint decide whether an edge node can serve models *elastically*—instantiated on demand and torn down when idle—or must pin one resident process per model. We measured both under strict subprocess-per-engine isolation, which is necessary because in a shared process only the first engine pays the one-time CUDA context-initialisation cost, and that asymmetry would contaminate any cross-engine comparison (it is exactly the 2.658 s artefact discussed in Appendix D).

The results support elastic deployment on both counts. Cold activation is *uniform* across all twelve engines (~ 0.18 – 0.22 s) and invariant to the distillation variant, and the full launch-to-ready wall time is ~ 0.65 s on the Jetson and ~ 1.0 s on the Legion—fast enough for on-demand instantiation, so a node need not keep every model resident and can reclaim its memory and power when idle. Peak memory is likewise modest and precision-ordered on the Jetson (189 MB at INT8 to 279 MB at FP16 of system RAM). On the 8 GB Orin Nano this is under 4% of system memory per model, leaving ample headroom to co-locate several distilled models—or other on-premise clinical software—on a single edge node. As with every cross-device figure, peak memory is *not* comparable across devices: the Legion figure reflects GPU memory, the Jetson figure shared system RAM.

C. What this jointly enables

Taken together, the frontier and the cold-start/memory results substantiate the “one edge node, many diagnostic models” picture stated in the Conclusion. The distilled student is not only efficient per inference but cheap to start and cheap to hold in memory, and the accuracy–energy and latency–throughput axes are both traversable without touching the weights. A single Orin Nano can therefore host several distilled models, or select an operating point per request, on fixed hardware—which is what makes the pipeline scalable in a practical, and not merely rhetorical, sense.

APPENDIX F

THERMAL STABILITY UNDER SUSTAINED LOAD

Real-time clinical operation means continuous, not bursty, inference, so a throughput figure measured over a few seconds is not sufficient evidence of deployability. Table IX and Fig. 7 report a 600 s sustained-inference soak on the Logit-KD engine at FP16 and INT8, with per-second telemetry (tegrastats on the Jetson, `nvidia-smi` on the Legion).

Neither device throttles. Throughput retention from start to steady state is $\geq 99.4\%$ in all four conditions (99.4–99.5% on the Legion, 99.8% on the Jetson), and clocks stay flat throughout. The Legion stabilises warmer ($\sim 60^\circ\text{C}$ steady, 65°C peak) while the Jetson remains markedly cooler ($\sim 45^\circ\text{C}$)—unsurprising given its far smaller thermal load—but the operative conclusion is the same for both: sustained real-time operation is thermally safe without active intervention or duty-cycling.

One methodological note. Retention here is the ratio of steady-state to initial throughput *within* the soak run, and absolute throughput in this table is *not* comparable to the batch-1 figures of Table VII, which come from a different measurement harness. The soak deliberately loads the device continuously while running concurrent per-second telemetry; its purpose is to detect thermal throttling, for which the retention ratio—not the absolute number—is the relevant quantity. As with all cross-device figures, steady-state power is reported per device and its two columns are not comparable to each other.

TABLE IX
SUSTAINED-LOAD THERMAL BEHAVIOUR (LOGIT-KD, 600 S SINGLE-STREAM). NEITHER PLATFORM THROTTLES. POWER DOMAINS ARE NOT DIRECTLY COMPARABLE (LEGION GPU-PACKAGE VERSUS JETSON BOARD VDD_IN).

Device	Prec.	T_{start} ($^\circ\text{C}$)	T_{steady} ($^\circ\text{C}$)	T_{max} ($^\circ\text{C}$)	Power (W)	Thrpt. ret. (%)
Legion	FP16	56	60	65	69.0	99.4
	INT8	58	60	64	69.3	99.5
Jetson	FP16	44	46	46	4.5	99.8
	INT8	45	45	45	3.9	99.8

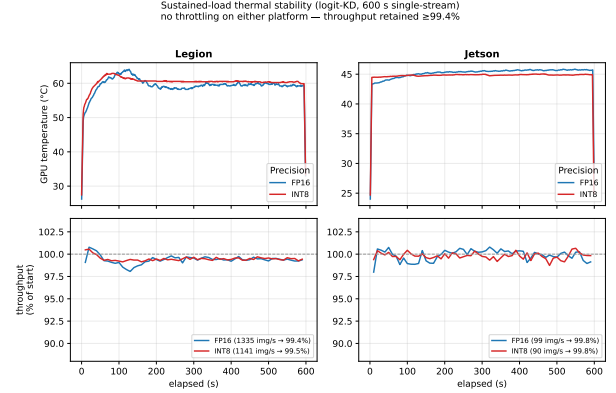


Fig. 7. Thermal soak: temperature and throughput over 600 s of sustained inference. Neither platform throttles.

APPENDIX G

FULL ROBUSTNESS AND OUT-OF-DISTRIBUTION RESULTS

A. Engine-level blur robustness

Table X reports the engine-level blur study underlying Fig. 3. AUD is the area under the degradation curve over $\sigma \in \{0, \dots, 4\}$ (higher = less robust); retention is $\text{AUROC}(\sigma=4)/\text{AUROC}(\sigma=0)$. The curves replicate across devices to within $\Delta < 0.003$, which is itself a useful sanity check: the phenomenon is a property of the trained weights, not of a particular TensorRT build.

We restrict the analysis to the three variants that span the distinct distillation paradigms—no distillation (Baseline), logit-level (Logit-KD) and feature-level (Feature-KD). This is precisely the contrast required to test whether the two robustness mechanisms dissociate, since it isolates the effect of the supervision *type*; Feature+Hard-KD is a sub-variant of the feature-level approach and would not introduce a new mechanism.

The numbers make the dissociation concrete. At FP16 the Baseline retains $\approx 66\%$ of its clean AUROC at $\sigma = 4$; Feature-KD retains $\approx 79\%$; and Logit-KD—despite having the *highest* clean score and the best INT8 tolerance—retains only $\approx 54\%$, i.e. it is more blur-fragile than the undistilled model it beats on every clean metric. This is not a bug in the evaluation but the expected signature of two different supervision signals: feature alignment to the ViT’s 14×14 token grid installs spatial structure that survives the removal of high-frequency texture,

whereas soft-logit supervision reshapes the output distribution and confers nothing spatial.

TABLE X
ENGINE-LEVEL INPUT ROBUSTNESS TO GAUSSIAN BLUR (MACRO(5) AUROC). AUD = AREA UNDER THE DEGRADATION CURVE OVER $\sigma \in \{0, \dots, 4\}$; LOWER IS MORE ROBUST.

Device	Variant	Prec.	Clean	$\sigma=2$	$\sigma=4$	AUD↓
Legion	Baseline	FP16	0.795	0.598	0.528	0.616
	Baseline	INT8	0.728	0.607	0.541	0.430
	Logit-KD	FP16	0.865	0.677	0.469	0.694
	Logit-KD	INT8	0.830	0.711	0.533	0.510
	Feature-KD	FP16	0.858	0.804	0.677	0.257
	Feature-KD	INT8	0.786	0.757	0.655	0.200
Jetson	Baseline	FP16	0.795	0.598	0.527	0.619
	Baseline	INT8	0.739	0.605	0.556	0.456
	Logit-KD	FP16	0.865	0.677	0.469	0.693
	Logit-KD	INT8	0.833	0.703	0.533	0.518
	Feature-KD	FP16	0.858	0.804	0.678	0.255
	Feature-KD	INT8	0.793	0.763	0.660	0.205

B. A caveat on reading retention

INT8 retention values are numerically *higher* than FP16 for every variant (Feature-KD: $\approx 83\%$ versus 79%). This must not be read as INT8 making a model more robust. It is largely a denominator effect: INT8 depresses the clean score, so a given absolute drop yields a higher retention ratio. Retention must therefore always be read alongside the absolute AUD, and any per- σ retention exceeding 100% reflects small-sample noise on the 234-image validation set rather than a genuine robustness gain.

C. Model-level corruption curves and domain shift

Figs. 8 and 9 show the complementary *model-level* (PyTorch checkpoint) degradation curves under blur and additive noise. They probe a different object from Table X—the trained network rather than the compiled engine—and are therefore a genuine cross-check rather than a repetition. The ordering is consistent across both: the greater the ViT’s influence on the student, the more slowly it degrades. This is the expected inheritance, given that conventionally trained CNNs base their decisions largely on texture and lose that evidence first [28].

Domain shift tests the same hypothesis with real, rather than synthetic, corruption. The NIH ChestX-ray dataset [4] was acquired with different X-ray equipment from CheXpert, so a model that has learned hardware-specific texture statistics rather than anatomy will drop sharply. Table XI reports the generalisation gap over the eight overlapping classes, and the pattern established above reappears exactly: the Baseline retains 89.79% of its in-distribution AUROC, the feature+logit student retains 93.66%—the best of the four—and the teacher itself, unsurprisingly, retains the most at 95.42%. Robustness to domain shift, like robustness to blur, is inherited in proportion to the amount of ViT supervision received.

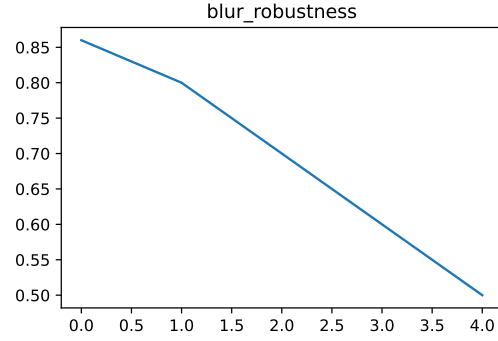


Fig. 8. Model-level robustness across increasing blur values (PyTorch checkpoints).

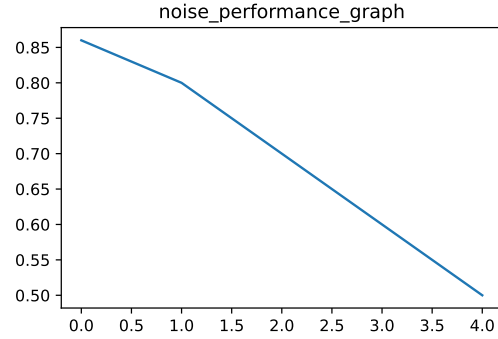


Fig. 9. Model-level robustness across increasing noise standard deviations (PyTorch checkpoints).

TABLE XI
GENERALISATION GAP ON NIH CHESTX-RAY (8 OVERLAPPING CLASSES). RETENTION INCREASES WITH THE AMOUNT OF ViT SUPERVISION RECEIVED.

Model	ID (CheXpert)	OOD (NIH)	Abs. drop	Retention
Baseline	0.7468	0.6705	−0.0762	89.79%
Feature-only-KD	0.8238	0.7500	−0.0737	91.05%
Feature+Logit-KD	0.8181	0.7662	−0.0519	93.66%
Logit-KD	0.8176	0.7396	−0.0780	90.46%
ViT teacher	0.8386	0.8001	−0.0384	95.42%

APPENDIX H CALIBRATION, OPERATING POINTS AND INTERPRETABILITY DETAILS

A. Reliability diagrams

Fig. 10 shows the reliability diagrams underlying Table III, at FP16 and INT8 on the Jetson. A perfectly calibrated model would lie on the diagonal; deviation above it indicates under-confidence, below it over-confidence. Two things are visible that the scalar ECE hides.

At FP16, the Baseline’s curve sags below the diagonal in the mid-confidence bins: it is systematically over-confident precisely in the region where a screening threshold would sit, which is the worst place for a miscalibrated model to be. The soft-target students (Logit-KD, Feature-KD) pull these

bins back towards the diagonal, which is what the halving of ECE ($0.123 \rightarrow 0.084/0.081$) actually means in operational terms. Feature+Hard-KD does not, confirming that the effect is carried by the teacher’s soft targets and not by feature alignment.

At INT8, the diagrams separate the variants sharply. Logit-KD’s curve degrades gracefully and stays near the diagonal (ECE $0.084 \rightarrow 0.119$), whereas Feature-KD’s collapses (ECE $0.081 \rightarrow 0.186$)—its probability mass is pushed into a few coarse quantization bins, breaking the correspondence between confidence and accuracy. This is the calibration-domain image of the AUROC result in Sec. V-B, and it strengthens the conclusion: soft-logit supervision produces output distributions that survive 8-bit quantization along *both* the ranking and the probability axes, while feature-based supervision survives along neither. The Baseline’s ECE shifting slightly under INT8 ($0.123 \rightarrow 0.107$) is incidental quantization noise and we do not over-interpret it.

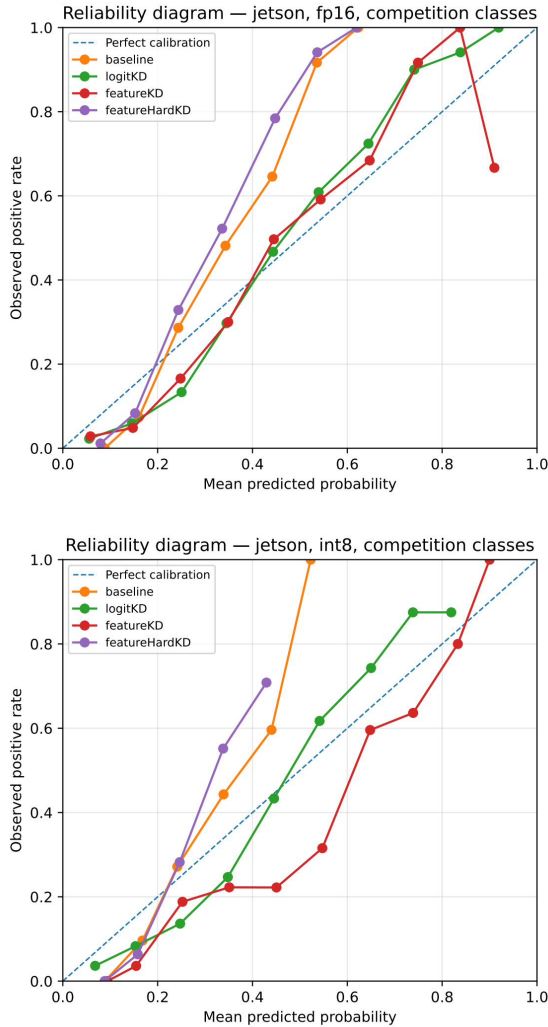


Fig. 10. Reliability diagrams on the Jetson at FP16 (top) and INT8 (bottom), ten equal-width probability bins, averaged over the five competition classes.

B. Per-class operating points

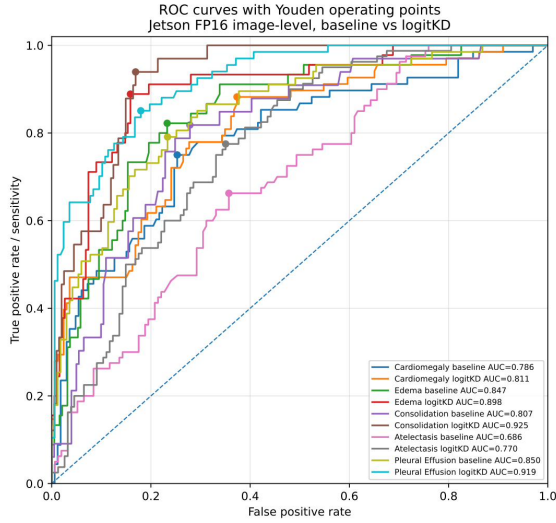
Whereas AUROC is threshold-independent, deployment requires a fixed decision threshold per pathology. Table XII gives the per-class sensitivity and specificity at the Youden-optimal threshold summarised in Table III, and Fig. 11 shows the corresponding ROC curves for Baseline versus Logit-KD.

The headline is that Logit-KD raises sensitivity—equivalently, reduces the false-negative rate—on *all five* findings, and that mean specificity rises as well, so the additional detections are not bought with false alarms. In a screening setting, where the dominant risk is failing to flag genuine pathology, this moves the model in the clinically preferred direction on both axes at once.

Cardiomegaly is the single exception and we analyse it rather than bury it. Its sensitivity rises from 0.750 to 0.882, but specificity falls from 0.747 to 0.627. With only 68 positives and a small nominal AUROC gain ($0.786 \rightarrow 0.811$), the paired bootstrap is underpowered and the confidence interval straddles zero; the two ROC curves are near-equivalent, so the Youden-optimal threshold is weakly determined. Logit-KD’s lower threshold ($0.223 \rightarrow 0.190$) therefore trades specificity for sensitivity rather than reflecting any genuine loss of discriminative power. For a sensitivity-prioritising screening use case this trade is defensible; we state it explicitly rather than presenting a uniform improvement.

TABLE XII
OPERATING POINTS AT YODEN’S J (JETSON, FP16). SENSITIVITY (= 1 – FALSE-NEGATIVE RATE) AND SPECIFICITY FOR THE FIVE COMPETITION CLASSES.

Pathology	Baseline		Logit-KD	
	Sens.	Spec.	Sens.	Spec.
Cardiomegaly	0.750	0.747	0.882	0.627
Edema	0.822	0.767	0.889	0.841
Consolidation	0.818	0.721	0.939	0.831
Atelectasis	0.663	0.643	0.775	0.649
Pleural Effusion	0.791	0.766	0.851	0.820
Mean	0.769	0.729	0.867	0.754



Operating points are computed at image level. Patient-cluster bootstrap is used only for AUROC CIs in Tasks 1-2.

Fig. 11. ROC curves, Baseline versus Logit-KD (Jetson, FP16). The curves are near-equivalent for Cardiomegaly, which is why its Youden threshold is weakly determined.

C. On the limits of the Grad-CAM comparison

Finally, we restate the epistemic status of Fig. 4. The teacher-versus-student panel is included to expose the cross-architecture localisation gap—the ViT aggregates evidence through global self-attention from its first layer onward, at 14×14 token resolution, while the student builds context hierarchically through local receptive fields and is read out at 7×7 —and *not* as evidence that spatial attention transfers. The two architectures localise evidence through mechanisms too different for such a comparison to be fair, and we make no claim of attention alignment; what the quantitative results establish is transferred is predictive behaviour at the logit level.

The student-versus-student comparison is the fair and publishable one: identical architecture, normalisation, target layer and display pixels mean that any difference in saliency is attributable to the distillation signal alone. Read anatomically, the distilled students place saliency on the lower lung zones and costophrenic angles—where effusion actually manifests—more consistently than the Baseline, whose maps are diffuse and more often stray from the pleural base. This is congruent with, but not proof of, the sensitivity gains reported above. Without region-level annotations we report no quantitative localisation score, and we therefore treat the entire interpretability analysis as illustrative.